MACHINE LEARNING and DEEP LEARNING

1. Consider a linear approach for modelling the relationship between variables: location of the land and price values in a house price data. What is the relationship if the dependent variable increases on the Y-axis and the independent variable increases on the X-axis?

a) Negative linear

b) Positive linear

c) Nonlinear

d) Bilateral

2. Consider the scenario of predicting if a child will pass or fail the examination. Which of the following algorithms is more suitable for the given task?

a) Linear regression

b) Multiple linear regression

c) Logistic regression

d) Polynomial regression

3. Assume that you have created 2 regression models M1 and M2 for predicting the temperature of the day time. Which regression line is the best-fit line among M1 and M2?

a) Line with the sum of the square of residuals is the minimum

b) Line with the sum of the square of residuals is the maximum

c) Line with the sum of residuals is the minimum

d) Line with the sum of the absolute value of residuals maximum

4. The students have developed a prediction model for predicting the weight of newborn infants based on the mother's medical history. Suggest appropriate metrics for evaluating regression models.

a) Recall

b) Precision

c) F-score

d) R squared

5. Consider a simple linear regression model Y=55.1+0.28X. What will the change in Y if X is varied or changed by 1 unit?

a) Y changes by 1 unit

b) no change in Y

c) Y changes by slope

d) Y changes by intercept

6. Assume that there is a very strong correlation between two variables age and date_of_birth in analyzing the covid data, then the correlation coefficient must be

a) any value larger than 1

b) much larger than 0, if the correlation is positive

c) much smaller than 0 if the correlation is negative

d) much larger than 0, if the correlation is negative

7. Which of the following is not a required assumption about the error term ε in the least squares regression?

a) The variance of the error term is the same for all values of x

b) The expected value of the error term is one

c) The values of the error term are independent

d) The error term is normally distributed

8. A ML engineer has carried out a regression analysis for predicting the salary of an employee with respect to his years of experience. But now his team leader has suggested to swap the explanatory and the response variables. What is the status of the updated regression model to the data?

a) the value of the correlation coefficient will change

b) the value of the sum of squared errors will change

c) the value of the coefficient of determination will change

d) the sign of the slope will change

9. Consider for a multiple regression model, the sum of the squared total is 400, and the sum of squared error is 75. Compute the multiple coefficients of determination.

a) 0.75

b) 0.81

c) 1.18

d) - 1.28

10. Assume a logistic regression model for predicting the forest fire. Odds are the ratio between probabilities(P) of the events (E) of forest fire .How will you calculate the odd ratio?

a) P of an E favourable to the outcome and P of E against the same outcome.

b) P of an E occurring.

c) P of an E not occurring.

d) P of E against the same outcome and P of an E favourable to the outcome

11. Assume 0.75 is the proportion of customers who gave a positive review for latest mobile model in the sample of online shopping customers. Compute the odd ratio of recommending the mobile.

a) 4

b) 3

c) 1

d) 0

12. Given a kNN classifier for analysing the quality of fabric into various categories, which one of the following statements is true?

a) Does not make any assumption on fabric data.

b) Increase in accuracy with an increase in the number of attributes

c) Learn from the training set immediately

d) can be used for classification only

13. A data scientist builds a KNN model for predicting the stage level of kidney disease. But data has noise. Which of the following ideas may improve the performance of the model?

a) Decrease the value of k

b) Changing value of k will not change the effect of the noise

c) Increase the value of k

d) increase the size of the data

14. A training data of DNA genes is given to the kNN algorithm, and the value of k < size of the training set. Assume that new test data is given to the model. How will the algorithm predict the class of a test example?

a) The most frequent class among the classes of k closest training examples

b) The most unique class among the classes of k closest training examples

c) The most unique class among the classes of closest training examples

d) The most unique class among the classes of closest training examples

15. The training data for gene sequences classification contains many attributes (categorical and continuous) with a lot of missing values. Which algorithm can be used for imputing missing values?

a) CNN

b) KNN

c) Linear Regression

d) Logistic regression

16. The data for predicting the class of DNA in the DNA sequence dataset contains attributes of categorical type only. Which of the distance measures can be used by the KNN algorithm for calculating the closeness?

a) Minkowski

b) Hamming Distance

c) Euclidean Distance

d) Manhattan Distance

17. The data scientist at Oceanware Corporation identifies the K means algorithm to be more appropriate for underwater creatures grouping. But then, they examine the data as of categorical type. Which of the following clustering algorithms can be used as an alternative to K-means for handling categorical data?

a) DBScan

b) Hierarchical

c) Density based

d) Kmodes

18. On grouping data according to their similarities, which clustering algorithm assigns data points to the nearest cluster centroid and is more suitable for high-dimensional data?

a) DBSCAN

b) K-Means

c) Agglomerative

d) Single linkage

19. Consider the location of data points p1(1,1) and p2(5,4). Compute its Manhattan distance.

a) 5

b) 7

c) 4

d) 3

20. The data scientists at Ignova Corporation apply K means algorithm for grouping the satellite data to identify unknown bodies.

Find the optimal approach followed for choosing K value.

a) Elbow Method

b) Choose large value of k

c) Choose small value of k

d) Choose k=1

21. Consider the problem of determining if the post graduate student will opt for job or higher studies. Assume the data has n Boolean variables and a Boolean label. What is the number of parameters needed to represent a Naive Bayes classifier?

a) n + 1

b) n

c) 2n + 1

d) 2n

22. Assume Naive Bayes classifier is trained using infinite fabric quality training data that satisfies all of its modelling assumptions. Make your inference on training error and test error?

a) It may not achieve zero training error but will always achieve zero test error

b) It will always achieve zero training error but may not achieve zero test error.

c) It may not achieve either zero training error or zero test error

d) It will always achieve zero training error and zero test error.

23. The ML engineer implements his cardiac prediction model using the K means algorithm. He makes some inferences on K means based on his experience. Which of the inference is correct?

a) The K-means algorithm is sensitive to outliers.

b) The model will always give zero error on the training data.

c) K-means algorithm will definitely give the same clustering results for different initializations

d) The K-means algorithm can detect non-convex clusters

24. Consider the data from marine sensors for K means clustering. Assume two variables, x1 and x2 used for clustering and k=3.

a) If x1 and x2 have a correlation of 1, the cluster centroids will be in a straight line

b) If x1 and x2 have a correlation of 0, the cluster centroids will be in a straight line

c) If x1 and x2 have a correlation of 1, the cluster centroids will not be in a straight line

d) If x1 and x2 have a correlation of 0, there is no cluster centroids

25. Smith and Jessica fit a linear regression model on a REVIEW dataset. Smith uses the original dataset, whereas Jessica normalizes each column of data. They found that the sum of squared residuals is different from each other. Why?

a) The intercept is not scaled.

b) The outcome is measured on a different scale.

c) One of them must have made a coding mistake.

d) The sum of residuals should be compared rather than the squared sum.

26. Consider a Naïve Bayes classifier with 4 input variables x1, x2, x3, x4 and a Boolean output Y. How many parameters must be computed to train such a classifier?

a) 5

b) 9

c) 7

d) 4

27. Assume that you are working with an EM algorithm for a model with latent variables. You are asked to modify the algorithm so that it finds MAP estimates instead of maximum likelihood estimates. Which step do you need to modify?

a) Expectation

b) Maximization

c) No need to do any modification

d) Both the Expectation and Maximization steps

28. For a dataset of three points {(3, 1), (1, 1), (-1, -1)}, what are the values of w and b given by a hard-margin SVM?

a) w=1, b=0

b) w=1, b=1

c) w=0, b=1

d) w=0, b=0

29. Consider performing k-means clustering on a one-dimensional dataset containing six data points {2, 6, 10, 14, 18, 7}, using k = 2, and the initial cluster centers are c1=4 and c2=12. Find the cluster centers after the first step.

a) 4, 14

b) 5, 13

c) 5, 14

d) 4, 13

30. Consider three clusters A = {0, 2, 5}, B = {3, 10}, and C = {12}, which of the two clusters will be merged at the next iteration of Hierarchical clustering when using (i) single linkage, and (ii) max-complete linkage.

a) AB, BC

b) AB, AB

c) AC, BC

d) AB, AC

31. Considering Self-Organizing map clustering, which of the following statement is incorrect?

a) Clustering by SOMs is in principle similar to the k-means clustering

b) It doesn't involve neural networks

c) The data points are initially assigned to the nodes at random

d) It starts by defining a number of nodes

32. In terms of the bias-variance tradeoff, a 2-nearest neighbor classifier has _____ than a 4-nearest neighbor classifier.

a) Higher variance, lower bias

b) Lower variance, higher bias

c) Higher variance, higher bias

d) Lower variance, Lower bias

33. Which of the following statements are true about bagging?

a) We choose random subsamples of the input points with replacements.

b) In the case of RF, bagging never gives a lower training error than a conventional decision tree.

c) Bagging is ineffective with logistic regression.

d) The main objective of bagging is to reduce the bias.

34. Which of the following methods would achieve an optimal policy

a) Q-Learning to estimate Q(s, a)

b) Direct evaluation to estimate V(s)

c) Temporal Learning to estimate V(s)

d) TD(1) learning to estimate V(s)

35. In AdaBoost learning, when would the misclassification penalty for a particular training observation increase going from round t to t+1?

a) If it is classified incorrectly by the weak learner trained in round t

b) If it is classified incorrectly by the full ensemble trained up to round t

c) If the observation is classified incorrectly by a majority of the weak learners trained up to round t

d) If the observation is classifier correctly by the weak learner trained in round t

36. Which of the following is true for the kNN algorithm?

a) As 'k' increases, the bias usually increases.

b) kNN can't be used for both classification and regression.

c) The decision boundary looks smoother with smaller values of k.

d) As 'k' increases, the variance usually increases.

37. While performing the backpropagation in NN, the gradient flows backward through the sigmoid or tanh non-linearities. Which of the following characteristic is true?

a) It cannot change the sign

b) It always reduces the weights

c) The weights will always increase

d) The sign would be reversed

38. Which are all the associated weights will be updated in SOM training.

a) For all the neurons

b) Only for the winning neurons

c) Winning neurons and their neighbors

d) Only for the neighbors of the winning neurons

39. In AdaBoost learning, why do we abandon the weak learners with less than 0.5 accuracy?

a) To reduce the number of classifiers

b) Might end up with negative weights, not suitable for voting

c) The variance might increase

d) To reduce the training time